Dr. Rajesh Kumar

Assistant Professor & Head, Department of Computer Science & Applications

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Research interests

Primary: automata theory (pushdown and multi-pushdown systems, formal languages), computational complexity (Turing-machine complexity, complexity of language operations), and formal verification (context-, round-, and scope-bounded reachability). **Additional:** search-based software engineering, genetic algorithms and metaheuristics, artificial intelligence, machine learning, data analytics. **Current directions (ongoing, unpublished):** multi-pushdown systems and bounded reachability.

Education

2020	**Ph.D., Computer Science**, Kurukshetra University. Thesis: *Search Based Software Engineering Using Metaheuristics*.
2009	**Master of Computer Applications (MCA)**, Kurukshetra University.
2006	**B.Sc. (Non-Medical)**, Kurukshetra University.

Academic appointments

Jul 2014 – present	**Assistant Professor & Head of Department**, Department of Computer Science & Applications, CRM Jat College, Hisar.
2010 – 2014	**Assistant Professor**, Department of Computer Science & Applications (DCSA), Kurukshetra University — postgraduate teaching (MCA / M.Sc.).
2009 – 2010	**Assistant Professor**, University College, Kurukshetra (UCK), Kurukshetra University — undergraduate teaching.

Publications

Google Scholar (retrieved 9 November 2026): 75 citations, h-index 3, i10-index 2. Indexing is stated only where independently verified. Titles follow the publisher's record.

Journal articles

1. Rajesh Kumar, Anju Jain, Rakesh Kumar. “Complexity of Finite State Turing Machine with Other Domain.” *Computer Science and Information Technologies*, 7(2), 196–202, 2026. DOI: 10.11591/csit.v7i2.p196-202.
2. Rajesh Kumar. “One-Way Tape Determinsitic [sic] PDA.” *REDVET – Revista electrónica de Veterinaria*, 24(4), 531–534, 2023. DOI: 10.69980/redvet.v24i4.1366.
3. Rajesh Kumar, Rakesh Kumar. “Optimizing Requirement Analysis by the Use of Meta-Heuristic in Search Based Software Engineering.” *International Journal of Electrical and Computer Engineering (IJECE)*, 9(5), 4336–4343, 2019. DOI: 10.11591/ijece.v9i5.pp4336-4343. Indexed in Scopus.
4. Rajesh Kumar, Rakesh Kumar. “Efficient Encoding Scheme in Genetic Algorithm for Requirement Optimization.” *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, 8(8S3), 489–493, 2019. ISSN 2278-3075.
5. Rajesh Kumar, Manju. “Complexity of Σ*Ā and its Connection with Logic.” *International Journal of Soft Computing and Engineering (IJSCE)*, 7(3), 34–36, 2017.
6. Rajesh Kumar, Manju. “Complexity of Binary and Unary Operations on Regular Grammar.” *International Journal of Soft Computing and Engineering (IJSCE)*, 7(3), 42–45, 2017.
7. Rajesh Kumar, Manju. “Automatic Scanning and Parsing using LEX and YACC.” *International Journal of Computer Science and Mobile Computing (IJCSMC)*, 6(7), 77–82, 2017.
8. Rajesh Kumar. “Innovation for Digital India.” *International Journal of Computer Science and Mobile Computing (IJCSMC)*, 6(7), 173–177, 2017.
9. Rajesh Kumar. “Complexity Measure of Turing Machine.” *International Journal of Computer Science and Mobile Computing (IJCSMC)*, 4, 135–138, 2015.
10. Rajesh Kumar. “State Complexity of Unary and Binary Operations on Regular Languages.” *Advanced Research International Publication House*, 5, 209–212, 2015.

11. Rajesh Kumar, Rakesh Kumar, Girdhar Gopal. “Novel Crossover Operator for Genetic Algorithm for Permutation Problems.” *International Journal of Soft Computing and Engineering (IJSCE)*, 3(2), 252–258, 2013.
12. Rajesh Kumar, Rakesh Kumar, Girdhar Gopal. “Alpha Cut Based Novel Selection for Genetic Algorithm.” *International Journal of Computer Applications*, 70(21), 13–17, 2013.
13. Rajesh Kumar, Rakesh Kumar, Girdhar Gopal. “Hybridization in Genetic Algorithms.” *International Journal of Advanced Research in Computer Science and Software Engineering (IJARCSSE)*, 3(4), 403–409, 2013.
14. Rajesh Kumar, Rakesh Kumar, Sudhir Narula. “A Population Initialization Method by Memetic Algorithm.” *International Journal of Advanced Research in Computer Science and Software Engineering (IJARCSSE)*, 3(4), 519–523, 2013.

Conference papers

15. Rajesh Kumar. “Complexity Measure of Turing Machine in Various Aspect.” National Conference on Data-Driven AI, Future Computing and Emerging Academic Perspectives (NC DAIFC-2026), FGM Government College, Adampur; published in *International Journal of Research Culture Society (IJRCS)*, 10 (Special Issue 44), 36–39, 2026.

Books and book chapters

16. Rajesh Kumar. *Digital Efficiency Tool*. Bharat Publication, New Delhi, 2025.
17. Rajesh Kumar. *Office Tool*. Laxmi Book Depot, 2024.
18. Rajesh Kumar. *Mathematical Foundation – II*. Directorate of Distance Education, Kurukshetra University, 2016.
19. Rajesh Kumar. *Web Engineering*. Directorate of Distance Education, Kurukshetra University, 2015.
20. Rajesh Kumar. “Logical Organisation of Computer” (book chapter). Directorate of Distance Education, Kurukshetra University, 2014.

Teaching

Theory of Computation · Design & Analysis of Algorithms · Data Structures · Operating Systems · Database Management Systems · Linux & Shell Programming · Python Programming · Artificial Intelligence · Data Analytics using R · Web Engineering

Professional and institutional service

- Reviewer, *IAES International Journal of Robotics and Automation (IJRA)*, Institute of Advanced Engineering and Science (2026 – present)
- Head, Department of Computer Science & Applications, CRM Jat College (current)
- Science Coordinator, National Education Policy (NEP) Committee · NAAC Coordinator · Examination Superintendent · Coordinator, Science Day / Technical Events
- Member, Organising Committees: “Cyber Terrorism and National Security” (Apr 2015); “Biodiversity & Sustainable Development” (Mar 2015); “Biodiversity & Environmental Sustainability in the Modern Era” (Feb 2019)

Invited and extension lectures

- Resource Person — four lectures, UGC-HRDC Refresher Course in Information Technology (Interdisciplinary), DCSA, Kurukshetra University (Nov 2019)
- Guest Lecture, “Data Structures & Algorithms” — DCSA, Kurukshetra University (Oct 2019)
- Extension Lectures, “Theory of Computation” — MCA, DCSA, Kurukshetra University (Sep 2018)
- Extension Lecture, “Artificial Intelligence” — Guru Nanak Khalsa College, Karnal (Feb 2017)
- Invited Lecture, “Open Source Software & MOOC” — Chhaju Ram College of Education, Hisar (Nov 2016)
- Extension Lecture, “Analysis & Design of Algorithms” — Guru Nanak Khalsa College, Karnal (Oct 2015)

Orientation and refresher courses

- Interdisciplinary Refresher Course, Advanced Research Methodology — TLC Ramanujan College, University of Delhi (May–Jun 2023), Grade A+
- UGC Refresher Course, Information Technology (Interdisciplinary) — UGC-HRDC, Kurukshetra University (Oct–Nov 2017), Grade A
- UGC Orientation Programme — UGC Academic Staff College, Kurukshetra University (Nov–Dec 2014), Grade A

Papers presented at seminars and conferences

- “Behaviour of Recursive Function in Lambda Calculus” — National Seminar, Government College Hansi (Jan 2026)

- "Complexity Measure of Turing Machine in Various Aspect" — National Conference DAIFC-26, FGM Government College, Adampur (Jan 2026)
- "Use of Automata Theory for Software Optimisation" — National Seminar, SD College, Ambala Cantt (Mar 2022)
- "Blended Approach to Learning — Its Challenges and Future" — National Seminar, SD College, Ambala Cantt (Jun 2021)
- "Solving of Traffic Issues using ICT" — National Seminar, CISKMV, Fatehpur-Pundri (Oct 2018)
- "Digital Addiction to Gadgets & its Impact on Health and Lifestyle" — National Seminar, CISKMV, Fatehpur-Pundri (Mar 2018)
- "Complexity of Σ*Ā and its Connection with Logic" — National Seminar, GMN College, Ambala Cantt (Mar 2016)
- "Innovations for Digital India" — UGC National Seminar, Dept. of Commerce, Kurukshetra University (Feb 2016)
- "Impact of Media Technologies on Personal & Social Life" — ICSSR National Seminar, CRM Jat College, Hisar (Apr 2015)
- Paper presented — ICSSR International Conference on Crop Technology-Led Agricultural Development, University College, Kurukshetra (Mar 2015)
- "Threats & Social Networking Sites" (co-author) — National Seminar, CISKMV, Fatehpur-Pundri (Feb 2014)
- "Automatic Scanning and Parsing using LEX and YACC" — National Conference on Computational Sanskrit, SD College, Ambala Cantt (Dec 2013)

Faculty development programmes and workshops

- Short Term Course, AI, Machine Learning and Cyber Security — MMTTC, Punjabi University, Patiala (Aug 2026), Grade A+
- FDP, Implementation of NEP-2020 — STRIDE, IGNOU, New Delhi (Jun 2023), Grade A
- Workshop on Leadership Development — UGC-HRDC, Himachal Pradesh University (Oct 2021)
- Short Term Course, Advances in Biometrics and Image Processing — DCSA, KUK (Mar 2021)
- Workshop on Intellectual Property Rights — DCSA, KUK (Mar 2021)
- FDP on E-Content Development — GAD-TLC with SD College, Ambala Cantt (Jul 2020), Grade B
- FDP on Artificial Intelligence — Sridevi Women's Engineering College, Hyderabad (May 2020)
- FDP on Intellectual Property Rights — S.A. Jain (PG) College, Ambala City (May 2020)
- FDP on Entrepreneurship Development — DCSA, KUK (Sep 2019)
- Workshop, NAAC Awareness — RUSA / IQAC, KUK (Jun 2019)
- RAITECOM-2019: Recent Advancements in IT & E-Commerce (2019)
- Recent Trends in Computer Applications and IT (2019)
- Two-Week FDP on Entrepreneurship — ABES / DST, at KUK (Nov–Dec 2018)
- Workshop on Data Analytics — DCSA, KUK (Sep 2015)
- National Seminar, Latest Developments in ICT — DMM, Kurukshetra (Feb 2015)
- NMEICT Awareness Programme — NIT Kurukshetra & NITTTR Chandigarh (Sep 2014)
- IBM SEED Program — Rational Software Architect, Kurukshetra (May 2014)
- National Workshop on Curriculum Revision — DCSA, KUK (Nov 2013)
- Workshop on National e-Governance Plan — IT Cell, KUK (Apr 2013)

Software development

Through Aryan TechLabs, which I founded, I design and deploy institutional software, including a college ERP (NEP/CBCS), a university employees management system for Haryana state universities, a college admissions portal and CRM, a student attendance system, the PraveshPath admissions eligibility matcher, and a Motor Accident Claims Tribunal case manager.

Technical skills

Systems	Linux administration, shell programming, server administration and automation (core expertise)
Research computing	LaTeX, academic documentation, MATLAB, SPSS (core expertise: LaTeX)
Languages	Python, TypeScript, JavaScript, C/C++, Java, R, Shell
Web & mobile	Node.js, NestJS, Express, Django REST Framework, Next.js, React, Tailwind CSS, React Native (Expo)
Data & DevOps	PostgreSQL, MySQL, Prisma ORM, Git/GitHub, CI/CD, Docker, Railway